Autodesk®
Scaleform®

DrawText API

Scaleform 4.2 DrawText API API GFx::Movie ActionScript
C++

:
: 2.0
: 2010 7 30

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GAMEWARE

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Autodesk® Scaleform® 4.2

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1. はじめに

Scaleform SDK
Flash® UI
Scaleform

SWF/GFX
HTML

Scaleform Flash
Flash Studio
TextField
TextFormat ActionScript (AS)
API
C++ API

ActionScript

Flash UI C++
Scaleform 3D HUD

Scaleform SDK C++ DrawText API
Scaleform
API Scaleform Render::Renderer
GFx::Movie
ActionScript

DrawText API 2 SWF/GFX
1
2 SWF GFx::MovieDef
(GFx::Loader::CreateMovie)
GFx::MovieDef GFx::DrawTextManager

2 C++ GFx::DrawTextManager
GFx::DrawText

- *GFx::DrawTextManager* – DrawText
/

- *GFx::DrawText –*
  Flash HTML
  SetColor SetFont SetFontStyle

1.1. GFx::DrawTextManager

DrawTextManager DrawText
GFx::StateBag

DrawTextManager :

1. DrawTextManager();

(SetFontProvider)

:

```
Ptr<DrawTextManager> pdm = *new DrawTextManager();
```

2. DrawTextManager(MovieDef* pmovieDef);

GFx::MovieDef
DrawTextManager
MovieDef MovieDef
DrawTextManager (
DrawTextManager DrawText)

:

```
Ptr<GFx::MovieDef> pmovieDef = *mLoader.CreateMovie("drawtext_fonts.swf",
Loader::LoadAll);
Ptr<GFx::DrawTextManager> pDm2 = *new DrawTextManager(pmovieDef);
```

3. DrawTextManager(GFx::Loader* ploader);

GFx::Loader

:

```
GFx::Loader mLoader;
..
Ptr<GFx::DrawTextManager> pDm1 = *new DrawTextManager(&mLoader);
```

1.1.1. GFx::DrawText の作成

DrawTextManager DrawText
DrawTextManager 1 DrawText

- `DrawText* CreateText(const char* putf8Str, const RectF& viewRect, const TextParams* ptxtParams = NULL, unsigned depth = ~0u);`
- `DrawText* CreateText(const wchar_t* pwstr, const RectF& viewRect, const TextParams* ptxtParams = NULL, unsigned depth = ~0u);`
- `DrawText* CreateText(const String& str, const RectF& viewRect, const TextParams* ptxtParams = NULL, unsigned depth = ~0u);`

DrawText (
NULL UTF-8 NULL Unicode/UCS-2 Scaleform::String)
`putf8str` : NULL UTF-8
`pwstr` : NULL Unicode/UCS-2
`str` : Scaleform::String

`viewRect` : ( )
(BeginDisplay)

`ptxtParams` :
TextParams :

```
struct TextParams
{
    Color                 TextColor;
    DrawText::Alignment  HAlignment;
    DrawText::VAlignment VAlignment;
    DrawText::FontStyle  FontStyle;
    float                FontSize;
    String              FontName;
    bool                 Underline;
```

```
    bool                Multiline;
    bool                WordWrap;
};
```

TextColor：

HAlignment：DrawText::Align_Left（）DrawText::Align_Right、DrawText::Align_Center、DrawText::Align_Justify

VAlignment：DrawText::VAlign_Top（）DrawText::VAlign_Bottom DrawText::VAlign_Center

FontStyle：DrawText::Normal DrawText::Bold DrawText::Italic DrawText::BoldItalic（DrawText::ItalicBold）

FontSize：（）

FontName：“Arial” “Times New Roman” “Bold” “Italic” FontStyle

Underline：

Multiline：/

WordWrap：/

ptxtParams DrawTextManager

/：

```
void SetDefaultTextParams(const TextParams& params);
const TextParams& GetDefaultTextParams() const;
```

DrawTextManager CreateHtmlText CreateText HTML DrawText：

```
// 指定されたHTMLを使ってGFx::DrawTextオブジェクトを作成し初期化する。
DrawText* CreateHtmlText(const char* putf8Str, const RectF& viewRect,
                         const TextParams* ptxtParams = NULL, unsigned depth = ~0u);
DrawText* CreateHtmlText(const wchar_t* pwstr, const RectF& viewRect,
                         const TextParams* ptxtParams = NULL, unsigned depth = ~0u);
DrawText* CreateHtmlText(const String& str, const RectF& viewRect,
```

```
                        const TextParams* ptxtParams = NULL, unsigned depth =
~0u);

  :

    //                                  GFx::DrawText
    String str("String No 2");
    DrawTextManager::TextParams params;
    params.FontName   = "Symbol";
    params.FontSize   = 30;
    params.FontStyle  = DrawText::Italic;
    params.Multiline  = false;
    params.HAlignment = DrawText::Align_Right;
    params.VAlignment = DrawText::VAlign_Bottom;

    Ptr<DrawText> ptxt;
    ptxt = *pdm->CreateText(str, RectF(20, 300, 400, 700), &params);

    // HTML    GFx::DrawText
    Ptr<DrawText> ptxt;
    ptxt = *pdm->CreateHtmlText("<p><FONT size='20'>AB
                                 <b>singleline</b><i> CD</i>O",
                                RectF(20, 300, 400, 700));
```

1.1.2. テキストのレンダリング

GFx::Movie
DisplayHandle GFx::DrawTextManager
DrawTextManager::SetViewport Viewport GFx

1.1.3. テキスト範囲の計測

DrawTextManager
  :

```
    SizeF GetTextExtent(const char* putf8Str, float width = 0,
                        const TextParams* ptxtParams = 0);
    SizeF GetTextExtent(const wchar_t* pwstr, float width = 0,
                        const TextParams* ptxtParams = 0);
    SizeF GetTextExtent(const String& str, float width = 0,
                        const TextParams* ptxtParams = 0);

    SizeF GetHtmlTextExtent(const char* putf8Str, float width = 0,
```

```
                            const TextParams* ptxtParams = 0);
    SizeF GetHtmlTextExtent(const wchar_t* pwstr, float width = 0,
                            const TextParams* ptxtParams = 0);
    SizeF GetHtmlTextExtent(const String& str, float width = 0,
                            const TextParams* ptxtParams = 0);
```

GetTextExtent TextParams

GetHtmlTextExtent HTML TextParams

‘width’

ptxtParams

(SetDefaultTextParams / GetDefaultTextParams) HTML
ptxtParams
HTML ‘ptxtParams’

:

```
  //
  String str("String No 2");
  DrawTextManager::TextParams params;
  params.FontName   = "Symbol";
  params.FontSize   = 30;
  params.FontStyle  = DrawText::Italic;
  params.Multiline  = false;
  params.HAlignment = DrawText::Align_Right;
  params.VAlignment = DrawText::VAlign_Bottom;

  SizeF sz = pdm->GetTextExtent(str, 0, params);

  params.WordWrap  = true;
  params.Multiline = true;
  sz = pdm->GetTextExtent(str, 120, params);

  // HTML
  const wchar_t* html =
      L"<p><FONT size='20'>AB <b>singleline</b><i> CD</i>O";
  sz = pdm->GetHtmlTextExtent(html);

  sz = pdm->GetHtmlTextExtent(html, 150);
```

1.2. GFx::DrawText

DrawText HTML

1.2.1. プレーン テキスト メソッドの設定と取得

SetText UTF-8 UCS-2 Scaleform::String
‘lengthInBytes’ UTF-8
‘lengthInChars’
UTF-8 UCS-2 NULL

```
void SetText(const char* putf8Str, UPInt lengthInBytes = UPInt(-1));
void SetText(const wchar_t* pstr,  UPInt lengthInChars = UPInt(-1));
void SetText(const String& str);
```

GetText UTF-8
HTML HTML

```
String GetText() const;
```

1.2.2. HTML テキスト メソッドの設定と取得

SetHtmlText UTF-8 UCS-2 Scaleform::String HTML
HTML
‘lengthInBytes’ UTF-8 ‘lengthInChars’

UTF-8 UCS-2 NULL

```
void SetHtmlText(const char* putf8Str, UPInt lengthInBytes = UPInt(-1));
void SetHtmlText(const wchar_t* pstr,  UPInt lengthInChars = UPInt(-1));
void SetHtmlText(const String& str);
```

GetHtmlText HTML SetColor SetFont

HTML

```
    String GetHtmlText() const;
```

1.2.3. テキスト フォーマットを設定するメソッド

```
void SetColor(Color c, UPInt startPos = 0, UPInt endPos = UPInt(-1));
```

(R G B A) [startPos..endPos]
'startPos' 'endPos'

```
void SetFont (const char* pfontName, UPInt startPos = 0, UPInt endPos = UPInt(-1))
```

[startPos..endPos]
'startPos' 'endPos'

```
void SetFontSize(float fontSize, UPInt startPos = 0, UPInt endPos = UPInt(-1))
```

[startPos..endPos]
'startPos' 'endPos'

```
void SetFontStyle(FontStyle, UPInt startPos = 0, UPInt endPos = UPInt(-1));

    enum FontStyle
    {
        Normal,
        Bold,
        Italic,
        BoldItalic,
        ItalicBold = BoldItalic
    };
```

[startPos..endPos]
'startPos' 'endPos'

```
void SetUnderline(bool underline, UPInt startPos = 0, UPInt endPos = UPInt(-1))
```

[startPos..endPos]
'startPos' 'endPos'

```
void SetMultiline(bool multiline);
```

('multiline' true) (false)

```
bool IsMultiline() const;
```

'true' 'false'

```
void SetWordWrap(bool wordWrap);
```

/

```
bool Wrap() const;
```

1.2.4. テキストの配置メソッド

Alignment VAlignment :

```
    enum Alignment
    {
        Align_Left,
        Align_Default = Align_Left,
        Align_Right,
        Align_Center,
        Align_Justify
    };
    enum VAlignment
    {
        VAlign_Top,
        VAlign_Default = VAlign_Top,
        VAlign_Center,
        VAlign_Bottom
    };
```

:

```
void SetAlignment(Alignment);
                        (                    )

Alignment   GetAlignment() const;
                        (                    )

void SetVAlignment(VAlignment);
                        (                    )

VAlignment  GetVAlignment() const;
                        (                    )
```

1.2.5. テキストの位置と向きの変更

DrawText

```
void SetRect(const RectF& viewRect);
```

```
RectF  GetRect() const;

void SetMatrix(const Matrix& matrix);

const Matrix GetMatrix() const;

void SetCxform(const Cxform& cx);

const Cxform& GetCxform() const;
```

2. DrawText サンプル コードの概要

Scaleform DrawText API
DrawText API Scaleform SDK

DrawText API

FxPlayerApp

FxPlayerApp
FxPlayerAppBase

FxPlayerApp OnInit DrawText
OnUpdateFrame

```
class   FxPlayerApp : public FxPlayerAppBase
{
    public:
    FxPlayerApp();

    virtual bool            OnInit(Platform::ViewConfig& config);
    virtual void            OnUpdateFrame(bool needRepaint);

    Ptr<GFx::DrawTextManager> pDm1, pDm2;
    Ptr<GFx::DrawText>          ptxt11, ptxt12,  ptxt21, ptxt22, ptxtImg,
                                pblurTxt, pglowTxt, pdropShTxt, pblurglowTxt;
    float   Angle;
    UInt32  Color;
};
```

DrawText DrawTextManager 2

方法 1

DrawTextManager GFx::Loader

```
pDm1 = *new DrawTextManager(&mLoader);
```

GFx::FontProvider FontProvider FxPlayerAppBase::OnInit

```
pDm1->SetFontProvider(mLoader.GetFontProvider());
```

API HTML DrawText DrawText Scaleform::String

```
//
DrawTextManager::TextParams defParams =
     pDm1->GetDefaultTextParams();
defParams.TextColor = Color(0xF0, 0, 0, 0xFF); // red, alpha = 255
defParams.FontName  = "Arial";
defParams.FontSize  = 16;
pDm1->SetDefaultTextParams(defParams);
```

CreateText SetDefaultTextParams

```
ptxt11 = *pDm1->CreateText ("Plain text, red, Arial,
                             16pts",RectF(20, 20, 500, 400));
```

DrawTextManager GetTextExtent DrawText

```
// StringとTextParams
String str(
   "Scaleform is a light-weight high-performance rich media vector
    graphics and user interface (UI) engine.");
GFx::DrawTextManager::TextParams params;
params.FontName   = "Arial";
params.FontSize   = 14;
params.FontStyle  = DrawText::Italic;
params.Multiline  = true;
params.WordWrap   = true;
params.HAlignment = DrawText::Align_Justify;
sz = pDm1->GetTextExtent(str, 200, &params);
ptxt12 = *pDm1->CreateText(str, RectF(200, 300, sz), &params);
ptxt12->SetColor(Render::Color(0, 0, 255, 130), 0, 1);
```

DrawText

DrawText DrawText DisplayHandle RenderThread

```
Render::Size<unsigned> viewSize = GetViewSize();
Render::Viewport dmViewport(viewSize.Width, viewSize.Height,
        int(viewSize.Width * GetSafeArea().Width),
        int(viewSize.Height * GetSafeArea().Height),
        int(viewSize.Width - 2 * viewSize.Width * GetSafeArea().Width),
        int(viewSize.Height - 2 * viewSize.Height * GetSafeArea().Height));
pDm1->SetViewport(dmViewport);
pDm1->Capture();
pRenderThread->AddDisplayHandle(pDm1->GetDisplayHandle(),
                                FxRenderThread::DHCAT_Overlay, false);
```

2

GFx::MovieDef MovieDef

DrawTextManger MovieDef

SWF Scaleform

Scaleform Integration Tutorial ( )

```
Ptr<MovieDef> pmovieDef = *mLoader.CreateMovie("drawtext_fonts.swf",
                                               Loader::LoadAll);
pDm2 = *new DrawTextManager(pmovieDef);
```

HTML (

HTML)

DrawTextManager::GetHtmlExtent

```
// MovieDefのフォントを使って、HTMLテキストを作成
const wchar_t* html = L"<P>o123 <FONT FACE=\"Times New Roman\" SIZE
                       =\"140\">"L" <b><i><FONT
        COLOR='#3484AA'> </FONT></i>  </b>     !</FONT></P>"
        L"<P><FONT FACE='Arial Unicode MS'>Hàny ; 华语/華語</FONT></P>"
        L"<P><FONT FACE='Batang'>한국어/조선말</FONT></P>"
        L"<P><FONT FACE='Symbol'>Privet!</FONT></P>";

GFx::DrawTextManager::TextParams defParams2 = pDm2->GetDefaultTextParams();
defParams2.TextColor = Color(0xF0,0,0,0xFF); //red,alpha = 255
defParams2.Multiline = true;
defParams2.WordWrap  = false;
SizeF htmlSz = pDm2->GetHtmlTextExtent(HtmlText, 0, &defParams2);
ptxt22 = *pDm2->CreateHtmlText(HtmlText, RectF(00, 100, htmlSz), &defParams2);
```

( )

:

```
SizeF sz;
  sz = pDm2->GetTextExtent("Animated");
  ptxt21 = *pDm2->CreateText("Animated", RectF(600, 400, sz));
  ptxt21->SetColor(Render::Color(0, 0, 255, 255));
  Angle = 0;
```

:

```
  SizeF sz;
  Ptr<GFx::DrawText>  pglowTxt
  pglowTxt = *pDm2->CreateText("Glow", RectF(800, 350, SizeF(200, 220)));
  pglowTxt->SetColor(Render::Color(120, 30, 192, 255));
  pglowTxt->SetFontSize(72);
  GFx::DrawText::Filter glowF(GFx::DrawText::Filter_Glow);
  glowF.Glow.BlurX = glowF.Glow.BlurY = 2;
  glowF.Glow.Strength = 1000;
  glowF.Glow.Color = Render::Color(0, 0, 0, 255).ToColor32();
  pglowTxt->SetFilters(&glowF);
```

DrawText DrawText DrawText
DisplayHandle RenderThread

```
  pDm2->SetViewport(dmViewport);
  pDm2->Capture();
  pRenderThread->AddDisplayHandle(pDm2->GetDisplayHandle(),
                                  FxRenderThread::DHCAT_Overlay, false);
```

DrawText SetMatrix SetCxform
FxPlayerApp::OnUpdateFrame :

```
  void FxPlayerApp::OnUpdateFrame( bool needRepaint )
  {
      DrawText::Matrix txt21matrix;
      Angle += 1;
      RectF r = ptxt21->GetRect();
      txt21matrix.AppendTranslation(-r.x1, -r.y1);
      txt21matrix.AppendRotation(Angle*3.14159f / 180.f);
      txt21matrix.AppendScaling(2);
      txt21matrix.AppendTranslation(r.x1, r.y1);
      ptxt21->SetMatrix(txt21matrix);

      pDm2->Capture();

      FxPlayerAppBase::OnUpdateFrame(needRepaint);
  }
```

: Bin/Samples (
) (
Projects/Win32/Msvc90/Samples/DrawText Visual Studio 2008)
drawtext_fonts.swf

: